

The future of inclusive development with DPGs

Vinod Sankaranarayanan, Business Head for Public Goods and Author at Thoughtworks and Sathyan Sethumadhavan, AI/ML Consultant, and Technical Engineering Leader at Thoughtworks

The use of fintech-powered services has exploded in Asia-Pacific, with China and India leading the way. In India, the Unified Payments Interface (UPI) is a key driver of this growth. UPI is a digital public infrastructure that allows people to send and receive money instantly between bank accounts. It is not tied to any one company, so people can switch between apps without losing their financial history. India's UPI has been very successful, with usage sometimes tripling year-on-year. In June 2023, UPI facilitated 93 billion transactions worth Rs. 14 lakh crores across 458 banks. Aside from payments and financial transactions, India has effectively delivered other digital public services such as identity and authentication to every citizen. These systems have all handled operations at a massive scale. This blog explores how DPGs or Digital Public Goods like UPI have achieved this level of success and sustained it.

Leveraging AI engineering

An ML model needs to do inference in less than 20-30 milliseconds for every UPI transaction — a daunting feat. In this scenario, one of the most valuable use cases for AI is fraud prevention. The AI needs to be reliable for both UPI and NPCI — the National Payments Corporation of India — that operates and manages UPI. It is also needed across the entire financial services ecosystem, including banks, fintechs and payment intermediaries.

Embracing open source

Thoughtworks partnered with UIDAI, The Unique Identification Authority of India to create an AI/ML workbench that empowers UIDAI to train and deploy AI solutions to prevent fraud. This open source-led approach means it is easier than it otherwise would be to successfully replicate across geographies, use cases and environments in both the identity and financial services space.

Thoughtworks also worked with ONDC, the Open Network for Digital Commerce, to develop an open-source mobility reference app. ONDC is a network that allows businesses to build their own stores, design offerings and reach customers. Thoughtworks began working with ONDC on a product taxonomy for the network and moved ahead to working on the mobility solution. Thoughtworks' product management team is still collaborating with ONDC's business team to help activate India's ecosystem and onboard them to the network.

What's next: generative AI for inclusive development

Generative AI has seen widespread adoption since the



mainstreaming of GPT-3 through the ChatGPT platform. While much of the implementation has been experimental — memes and content summarization, the potential of generative AI is endless.

Thoughtworks' experience of working with the UIDAI and ONDC has presented specific and exciting opportunities to leverage AI in digital public goods. Here are a few examples of what the tech consultancy has been able to accomplish.

Being vernacular-friendly

Most large language models (LLMs) that power generative AI are built on data in the English language. This limits the tech's use for citizen-scale digital public goods in countries like India where there are several languages and dialects spoken (aside from the high degree of illiteracy in rural areas). To address this contingency, Thoughtworks created Jugalbandi, a platform that combines existing LLMs and translation models to power conversational AI.

One of the most successful applications of Jugalbandi is its use by the Indian government to provide information about government schemes to citizens on WhatsApp. Citizens can send a message in their preferred language and Jugalbandi will look through myscheme.gov.in, India's citizen engagement platform, and reply in the same language.

Gen AI-powered fraud detection

55% of online scams are linked to UPI. UPI, NPCI and its other ecosystem partners already use AI models to combat fraud. But generative AI could make it even more effective by taking large volumes of transaction data to analyze signals, predict the next steps and identify potential fraud.

Domain context for generative AI

LLMs are trained on petabytes of publicly available and generic data such as books and web pages. This approach is not suitable for domain-specific needs. Generative AI can be useful for real-world scenarios when it is 'trained for domain context.' This could include familiarizing the tech with terminology and usage for a specific domain or feeding it an organization's proprietary data for a more specific context. Bloomberg GPT is an excellent example.

In the Indian context, generative AI, trained on health-care data can analyze large volumes of (privacy and confidentiality-protected) medical reports from primary health centers to track/forecast common diseases. Similar use cases exist across various industries, user groups and government departments.

Conclusion

Thoughtworks believes in the power of digital public goods to improve citizens' lives and make governments more efficient. Digital public goods further a country's digital transformation efforts. They can help governments build better services, make better decisions and improve citizens' lives.

When DPGs combine their potential with the power of tech like generative AI — inclusive development could have no limits. What we've seen so far is only the beginning of what generative AI can do for businesses, not-for-profits and governmental entities. With thoughtful investment in infrastructure and applications, corporate India can build for the country and leverage existing digital public goods for the benefit of everyone.